Columbia Business School

B9339 Quantitative Hedge Fund Strategies
Professor: Achilles Venetoulias
Fall 2026

Homework Assignment #4

Due: Thursday, October 29, 2026; 2:20pm (EST)

- Homework assignments can be done in groups of three or fewer people. Only turn in one write-up per group.

- In your write-up, please:
  - Include (preferably on the first page) a summary of your answers to all the questions.
  - Give your answers to at least five digits (if appropriate) after the decimal place (this makes it easier to spot which answers are correct or where mistakes occur).

- In your solutions to the individual questions, please:
  - Explain the main steps or formulas used to arrive at your solution. Specifically, *show* and *explain* the main formulas used in any spreadsheet or software output you hand in. The formulas in the annotated output can be handwritten (do not have to be typeset).
  - Include row and column headings in spreadsheet printouts (in Excel, choose *File—Print Preview*, then click on *Setup*, then choose the *Sheet* tab and check the "row and column headings" box).

- Please organize and structure software output you hand in so that it is easy for others to understand. Spend extra time, after getting the answers correct, to create clean-looking, understandable submissions.

- It is not sufficient to turn in a spreadsheet or some other software output (in printout or electronic form) with only numbers showing. Points will be deducted for write-ups turned in without the main formulas shown and explained, even if the numerical solutions are correct. How much detail is needed in the explanations? Enough detail so that anyone else in the course (student, TA, instructor) could follow (and, relatively easily, reproduce) your work and final solutions using only your write-up.

- Late homework submissions will not be accepted.

1. (*Inverting returns*) We have discussed in class, in the context of an FX Carry strategy, the "care" that is required in computing currency returns accurately. The value of *Currency B* to *Currency A* is the reciprocal of the value of *Currency A* to *Currency B*. This reciprocity, however, does not hold for returns.
   a. Explain why the return of a currency pair (e.g. JPYUSD) for any period is not the inverse of the return of the same pair in the opposite direction (USDJPY) for the same period.
   b. Assuming you know the first two moments (expectation and variance) of the return distribution, but not the full distribution, of a currency pair in one direction (e.g. JPYUSD), derive approximate formulas for the expected return and variance of the return distribution of the same currency pair in the opposite direction (USDJPY) without computing the (unspecified) integrals involved.
   c. Use the data from Homework Assignment #3 ("hw3.data.FX.dly.xlsx") to construct (daily) return time-series for NZD/JPY and JPY/NZD, and apply your answers from (b) above. Do the empirical results agree with your derivations?

2. (*The 72 Rule*) Practitioners have used this rule of thumb for decades: with discrete compounding, an investment which grows at an interest rate of r percent per period requires $n \approx 72/(100 * r)$ periods in order to double in value.



Using both the exact value of $\ln(2)$ and the approximate value of $\ln(2) \approx 0.69$, give two approximations of your own (one better than the other) which are "close" to the "72 Rule."
   a. Using any software of your choice, assess the sensitivity of the (five) approximations to the value of the interest rate r.
   b. For what value of the interest rate is the "72 Rule" formula closest to the 'true' n?

3. (*FX Carry*) In the following excerpt from the Jurek, *Crash-neutral currency carry trades,* Journal of Financial Economics, **v113** (2014), pp. 325–347 paper, explain:

(a) The expression for q_p in Equation (3);
(b) Why the expression in Equation (4) "makes transparent that the return to the strategy is bounded from below";
(c) Why the expression in Equation (4) "makes transparent … that for terminal realizations of the exchange rate that are above the strike, K_p, the strategy payoff is steeper than in the unhedged carry trade, reflecting the hedging of the option overlay";
(d) The sentence "To obtain the return of the trade in which the option overlay is not hedged (or the strike of the put converges to 0), the hedge delta, δ_p, in the above formula should be set to zero."

To describe the composition of the hedged carry trade portfolio consider again the currency pair J/I, with J having the higher interest rate. The hedged portfolio involves a long position in currency J, funded in currency I, and a position in put options on the S_t^{ji} exchange rate. To protect against the risk of depreciation of currency J against currency I, assume the trader purchases FX puts with a strike price K_p at a cost of $\mathcal{P}_t(K_p, \tau)$ per put. For every q_p puts purchased, the trader must also purchase an additional $-q_p \cdot \delta_p$ units of the foreign currency to hedge the (negative) delta of the put options. Finally, I assume the purchase price of the puts is covered by borrowing additional funds in currency I. At time $t+1$ the return on this portfolio is given by:

$$\begin{aligned} \tilde{r}_{t+\tau}^{ji} &= \exp\left(y_{t,t+\tau}^{j} \cdot \tau\right) \cdot (1 - q_p \cdot \delta_p) \cdot \frac{S_{t+\tau}^{ji}}{S_t^{ji}} + q_p \cdot \max\left(\frac{K_p}{S_t^{ji}} - \frac{S_{t+\tau}^{ji}}{S_t^{ji}}, 0\right) - \\ &\quad - \exp\left(y_{t,t+\tau}^{i} \cdot \tau\right) \cdot \left((1 - q_p \cdot \delta_p) + q_p \cdot \frac{\mathcal{P}_t(K_p, \tau)}{S_t^{ji}}\right) \end{aligned} \tag{2}$$

In order to eliminate all currency exposure below the strike price, K_p, the quantity of puts must satisfy,

$$q_p = \exp\left(y_{t,t+\tau}^{j} \cdot \tau\right) \cdot (1 - q_p \cdot \delta_p) \quad \rightarrow \quad q_p = \frac{\exp\left(y_{t,t+\tau}^{j} \cdot \tau\right)}{1 + \exp\left(y_{t,t+\tau}^{j} \cdot \tau\right) \cdot \delta_p} \tag{3}$$

With the above quantity restriction, the return equation can be re-expressed as:

$$\begin{aligned} \tilde{r}_{t+\tau}^{ji} &= q_p \cdot \max\left(\frac{K_p}{S_t^{ji}}, \frac{S_{t+\tau}^{ji}}{S_t^{ji}}\right) - \exp\left(y_{t,t+\tau}^{i} \cdot \tau\right) \cdot \left((1 - q_p \cdot \delta_p) + q_p \cdot \frac{\mathcal{P}_t(K_p, \tau)}{S_t^{ji}}\right) \\ &= q_p \cdot \max\left(\frac{S_{t+\tau}^{ji} - K_p}{S_t^{ji}}, 0\right) - q_p \cdot \exp\left(y_{t,t+\tau}^{i} \cdot \tau\right) \cdot \frac{\mathcal{C}_t(K_p, \tau)}{S_t^{ji}} \end{aligned} \tag{4}$$

This expression makes transparent that the return to the strategy is bounded from below, and that for terminal realizations of the exchange rate that are above the strike, K_p, the strategy payoff response is steeper than in the unhedged carry trade, reflecting the hedging of the option overlay. Specifically, the upside sensitivity is magnified by a factor, $\frac{q_p}{\exp\left(y_{t,t+\tau}^{j} \cdot \tau\right)}$, which is strictly greater than one. To obtain the return of the trade in which the option overlay is not hedged (or the strike of the put converges to zero), the hedge delta, δ_p, in the above formula should be set to zero. As before, if i is not the investor's home currency, the above return needs to converted to the home currency, h, by multiplying it by $\frac{S_{t+\tau}^{ih}}{S_t^{ih}}$.